Complex Numbers

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Complex numbers

Definition 0.1 (Imaginary unit)

Definition 0.1 (Imaginary unit)**.** The **imaginary unit** i is a number whose square is -1:

$$i^2 \stackrel{\text{def}}{=} -1.$$

Apart from that one rule, i obeys the usual rules of arithmetic:

- sums and products with i are commutative;
- sums and products with i are associative;
- multiplication distributes over addition.

Axler (2024, Definition 1.1, p. 2) makes this idea rigorous: it defines a complex number as an ordered pair (a, b) of real numbers, written $a + bi$, defines addition and multiplication of such pairs, writes $0 + 1i$ as i, and leaves it to the reader to verify that $i^2 = -1$.

Example 0.1 (Powers of the imaginary unit)

Example 0.1 (Powers of the imaginary unit)**.** The powers of i repeat in a cycle of four:

$$\begin{aligned} i^3 &= i^2 \cdot i && \text{(split off one factor of } i\text{)} \\ &= (-1) \cdot i && \text{(Definition 0.1)} \\ &= -i && \text{(multiply)} \end{aligned}$$

and

$$\begin{aligned} i^4 &= i^2 \cdot i^2 && \text{(split the power into two squares)} \\ &= (-1) \cdot (-1) && \text{(Definition 0.1, twice)} \\ &= 1 && \text{(multiply)} \end{aligned}$$

so the cycle starts again:

$$\begin{aligned} i^5 &= i^4 \cdot i && \text{(split off one factor of } i\text{)} \\ &= 1 \cdot i && (i^4 = 1) \\ &= i && \text{(multiply)} \end{aligned}$$

and the powers run $i, -1, -i, 1, i, \ldots$.

Example 0.2 (No real number squares to -1)

Example 0.2 (No real number squares to -1)**.** The imaginary unit is not a real number, because the square of every real number x is at least 0:

- if $x \geq 0$, then $x^2 = x \cdot x$ is a product of two nonnegative numbers, so $x^2 \geq 0$;
- if $x < 0$, then $x^2 = x \cdot x$ is a product of two negative numbers, so $x^2 > 0$.

For instance, $3^2 = 9$ and $(-3)^2 = 9$; neither is -1. So the equation $x^2 = -1$ has no real solution, and Definition 0.1 introduces a new number to solve it.

Theorem 0.1 (The number $-i$ also squares to -1)

Theorem 0.1 (The number $-i$ also squares to -1)**.**

$$(-i)^2 = -1.$$

Proof

Proof.

$$\begin{aligned} (-i)^2 &= (-1)^2\, i^2 && (-i = (-1) \cdot i, \text{ and } (ab)^2 = a^2 b^2 \text{ because products commute and associate}) \\ &= 1 \cdot (-1) && ((-1)^2 = 1 \text{ and Definition 0.1}) \\ &= -1 && \text{(multiply)} \end{aligned}$$

□

Example 0.3 (Squaring i and $-i$ in R)

Example 0.3 (Squaring i and $-i$ in R)**.** R has the imaginary unit as `1i`. The code below

squares it and its negative.

```
squares <- c(i = (1i)^2, minus_i = (-1i)^2)
squares
#>       i minus_i
#> -1+0i -1+0i
```

Both squares have real part -1 and imaginary part 0, as Definition 0.1 and Theorem 0.1 say.

Remark 0.1 (The rule does not tell i from $-i$)

Remark 0.1 (The rule does not tell i from $-i$). Because $-i$ obeys the same rule as i (Theorem 0.1), the rule $i^2 = -1$ does not single one of them out. The symbol i names one of these two square roots of -1, chosen once and for all.

Definition 0.2 (Complex number)

Definition 0.2 (Complex number)**.** A **complex number** is a number of the form

$$z = a + b\,i,$$

where a and b are real numbers and i is the imaginary unit (Definition 0.1).

- The **real part** of z is $\operatorname{Re} z \stackrel{\text{def}}{=} a$.
- The **imaginary part** of z is $\operatorname{Im} z \stackrel{\text{def}}{=} b$.
- The set of all complex numbers is $\mathbb{C} \stackrel{\text{def}}{=} \{a + b\,i : a, b \in \mathbb{R}\}$.

Two complex numbers are equal exactly when their real parts are equal and their imaginary parts are equal.

Axler (2024, Definition 1.1, p. 2) defines $\mathbb{C}$ as the set of ordered pairs (a, b) of real numbers, written $a + bi$, and Axler (2024, Definition 4.1, p. 120) defines the real and imaginary parts.

Example 0.4 (Real and imaginary parts)

Example 0.4 (Real and imaginary parts)**.** For $z = 2 - 5\,i$, the real part is $\operatorname{Re} z = 2$ and the imaginary part is $\operatorname{Im} z = -5$: the imaginary part is the real number -5, not $-5\,i$.
The numbers $1 + i$ and $1 - i$ have the same real part, 1, but different imaginary parts, 1 and -1, so they are different complex numbers.
The real number 7 is the complex number $7 + 0\,i$, with real part 7 and imaginary part 0.

Theorem 0.2 (Every real number is a complex number)

Theorem 0.2 (Every real number is a complex number)**.** *A complex number z is a real number exactly when* $\operatorname{Im} z = 0$*. In particular, every real number a is the complex number $a + 0\,i$, with* $\operatorname{Re} a = a$ *and* $\operatorname{Im} a = 0$*, so* $\mathbb{R}$ *is a subset of* $\mathbb{C}$*.*

Proof

Proof. Let a be a real number. Since $0\,i = 0$,

$$\begin{aligned} a + 0\,i &= a + 0 && (0\,i = 0) \\ &= a && \text{(Theorem 4 in Convexity, Infimum and Sums)} \end{aligned}$$

so a is the complex number $a + 0\,i$, whose imaginary part is 0. The real part of $a + 0\,i$ is a, and by the equality rule in Definition 0.2 no other pair of real and imaginary parts gives the same number.
Conversely, if $z = a + b\,i$ has $b = \operatorname{Im} z = 0$, then $z = a + 0\,i = a$ by the same calculation, so z is real. □

Example 0.5 (Which complex numbers are real)

Example 0.5 (Which complex numbers are real)**.** R stores complex numbers as a real part and an imaginary part. The code below converts the real number 7 to a complex number and reads the parts of three complex numbers.

```
z <- c(as.complex(7), 2 - 5i, 3 + 0i)
data.frame(z = z, re = Re(z), im = Im(z), is_real = Im(z) == 0)
#>      z re im is_real
#> 1 7+0i  7  0    TRUE
#> 2 2-5i  2 -5   FALSE
#> 3 3+0i  3  0    TRUE
```

The real number 7 becomes the complex number 7+0i, with imaginary part 0. Of the 3 numbers, 2 have imaginary part 0, and by Theorem 0.2 these are the real ones. The number $2 - 5\,i$ is not real, because its imaginary part is -5.

Theorem 0.3 (Adding and multiplying complex numbers)

Theorem 0.3 (Adding and multiplying complex numbers)**.** *For real numbers a, b, c, d:*

$$(a + b\,i) + (c + d\,i) = (a + c) + (b + d)\,i$$

and

$$(a + b\,i)(c + d\,i) = (ac - bd) + (ad + bc)\,i.$$

Proof

Proof. **Sum.**

$$\begin{aligned}(a + b\,i) + (c + d\,i) &= a + c + b\,i + d\,i && \text{(rearrange the terms)}\\ &= (a + c) + (b + d)\,i && \text{(factor out } i\text{)}\end{aligned}$$

Product.

$$\begin{aligned}(a + b\,i)(c + d\,i) &= ac + ad\,i + bc\,i + bd\,i^2 && \text{(distribute)}\\ &= ac + ad\,i + bc\,i - bd && \text{(Definition 0.1)}\\ &= (ac - bd) + (ad + bc)\,i && \text{(group the real terms and the terms with } i\text{)}\end{aligned}$$

□

Axler (2024, Definition 1.1, p. 2) takes these two formulas as the definitions of addition and multiplication.

Example 0.6 (Adding and multiplying two complex numbers)

Example 0.6 (Adding and multiplying two complex numbers)**.** Let $w = 1 + 2\,i$ and $z = 3 - i$. Their sum is

$$\begin{aligned}w + z &= (1 + 3) + (2 + (-1))\,i && \text{(Theorem 0.3, sum)}\\ &= 4 + i && \text{(add)}\end{aligned}$$

and their product, multiplying out directly, is

$$\begin{aligned}wz &= 1 \cdot 3 + 1 \cdot (-i) + 2\,i \cdot 3 + 2\,i \cdot (-i) && \text{(distribute)}\\ &= 3 - i + 6\,i - 2\,i^2 && \text{(multiply)}\\ &= 3 - i + 6\,i + 2 && \text{(Definition 0.1)}\\ &= (3 + 2) + (-1 + 6)\,i && \text{(group the real terms and the terms with } i\text{)}\\ &= 5 + 5\,i && \text{(add)}\end{aligned}$$

The product formula in Theorem 0.3 gives the same answer: with $a = 1$, $b = 2$, $c = 3$ and $d = -1$,

$$\begin{aligned}ac - bd &= 3 - (-2)\\ &= 5\end{aligned}$$

and

$$\begin{aligned} ad + bc &= -1 + 6 \\ &= 5. \end{aligned}$$

Definition 0.3 (Complex conjugate)

Definition 0.3 (Complex conjugate)**.** The **complex conjugate** of a complex number $z = a + b\,i$ (Definition 0.2) is

$$\overline{z} \stackrel{\text{def}}{=} a - b\,i.$$

Axler (2024, Definition 4.2, p. 120) gives the same definition, as $\overline{z} = \operatorname{Re} z - (\operatorname{Im} z)\,i$.

Example 0.7 (Complex conjugates)

Example 0.7 (Complex conjugates)**.**

- $\overline{3 + 4\,i} = 3 - 4\,i$.
- $$\begin{aligned} \overline{-2\,i} &= \overline{0 + (-2)\,i} \\ &= 0 - (-2)\,i \\ &= 2\,i. \end{aligned}$$
- $$\begin{aligned} \overline{5} &= \overline{5 + 0\,i} \\ &= 5 - 0\,i \\ &= 5 : \end{aligned}$$

  a real number is its own complex conjugate. A complex number with a nonzero imaginary part, such as $3 + 4\,i$, is not.

Definition 0.4 (Absolute value (modulus) of a complex number)

Definition 0.4 (Absolute value (modulus) of a complex number)**.** The **absolute value**, or **modulus**, of a complex number $z = a + b\,i$ (Definition 0.2) is

$$|z| \stackrel{\text{def}}{=} \sqrt{a^2 + b^2}.$$

Axler (2024, Definition 4.2, p. 120) gives the same definition.

Example 0.8 (Absolute values of complex numbers)

Example 0.8 (Absolute values of complex numbers)**.**

- $$\begin{aligned} |3+4\,i| &= \sqrt{3^2+4^2} \\ &= \sqrt{25} \\ &= 5. \end{aligned}$$
- $$\begin{aligned} |-2\,i| &= \sqrt{0^2+(-2)^2} \\ &= \sqrt{4} \\ &= 2. \end{aligned}$$
- For a real number $a = a + 0\,i$,

  $$\begin{aligned} |a| &= \sqrt{a^2+0^2} \\ &= \sqrt{a^2}, \end{aligned}$$

  which is the usual absolute value of a; for instance,

  $$\begin{aligned} |-3| &= \sqrt{9} \\ &= 3. \end{aligned}$$

Theorem 0.4 (A complex number times its conjugate)

Theorem 0.4 (A complex number times its conjugate)**.** *For every complex number z,*

$$z\,\overline{z} = |z|^2.$$

In particular, $z\,\overline{z}$ is a nonnegative real number.

Proof

Proof. Write $z = a + b\,i$ with a, b real. Then

$$
\begin{aligned}
z\,\overline{z} &= (a + b\,i)(a - b\,i) && \text{(Definition 0.3)} \\
&= a^2 - ab\,i + ab\,i - b^2\,i^2 && \text{(distribute)} \\
&= a^2 - b^2\,i^2 && \text{(cancel } ab\,i\text{)} \\
&= a^2 + b^2 && \text{(Definition 0.1)} \\
&= |z|^2 && \text{(Definition 0.4)}
\end{aligned}
$$

□

Axler (2024, result 4.4, p. 121) lists this identity among the properties of complex numbers.

Example 0.9 (Multiplying $3 + 4\,i$ by its conjugate)

Example 0.9 (Multiplying $3 + 4\,i$ by its conjugate)**.**

$$
\begin{aligned}
&(3 + 4\,i)(3 - 4\,i) \\
&= 9 - 12\,i + 12\,i - 16\,i^2 && \text{(distribute)} \\
&= 9 - 16\,i^2 && \text{(cancel } 12\,i\text{)} \\
&= 9 + 16 && \text{(Definition 0.1)} \\
&= 25 && \text{(add)}
\end{aligned}
$$

which is $|3 + 4\,i|^2 = 5^2$ from Example 0.8, as Theorem 0.4 says.

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